When Should Multi-Agent Systems Replace or Extend Classical Retrieval-Augmented Generation in Education?

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Abstract

This draft examines when multi-agent large language model systems should replace or extend classical retrieval-augmented generation (RAG) in educational settings. The argument is intentionally narrow: multi-agent systems are not presented as a universal substitute for retrieval, but as a potentially stronger orchestration pattern for educational tasks whose main difficulty lies in coordination rather than document access alone. The paper is framed as a taxonomy and evaluation-protocol blueprint rather than as a completed empirical study. It organizes educational tasks along two axes—knowledge grounding need and pedagogical coordination complexity—and uses that framing to compare classical RAG, agentic RAG, and non-RAG multi-agent systems. The proposed evaluation protocol emphasizes groundedness, pedagogical quality, adaptivity, robustness, cost, and latency under shared budgets and shared task conditions, while the benchmark mapping distinguishes education-facing evidence from transferable mechanism-level stress tests.

1 Introduction

Large language model systems for education are often framed as retrieval-augmented generation pipelines: a user question is matched to a document collection, relevant passages are retrieved, and a model produces a response grounded in those passages. This pattern is effective when the task is primarily factual, citation-sensitive, or directly answerable from a curated corpus. However, many educational tasks are not merely information-access problems. Tutoring, feedback, misconception diagnosis, study planning, rubric-based critique, and adaptive scaffolding often require decomposition, role specialization, iterative critique, and multi-turn pedagogical adaptation.

This draft therefore asks a narrower and more defensible question than whether multi-agent systems replace retrieval entirely. The central question is whether multi-agent systems can replace or extend *classical RAG as the top-level orchestration pattern* for educational tasks whose difficulty lies in pedagogical coordination rather than retrieval alone. In this framing, retrieval remains important for evidence-grounded tasks, but it may no longer be sufficient as the primary architectural abstraction when tasks require explicit negotiation among pedagogical roles, monitoring of learner state, or decomposition into multiple instructional subgoals.

The paper is organized around two dimensions: the degree of knowledge grounding required by a task and the degree of pedagogical coordination complexity required to solve it well. The central hypothesis is that classical RAG remains strong on relatively simple knowledge-grounded tasks, while agentic or multi-agent systems become more compelling as tasks demand adaptive sequencing, critique, or role-specialized reasoning. This perspective avoids universal replacement claims and instead emphasizes boundary conditions, mechanism, and fair architectural comparison.

In its current form, the paper should be read as a taxonomy and evaluation blueprint grounded in existing benchmarks, not as a report of completed empirical results.

The contributions of the paper are threefold. First, it proposes a task taxonomy that separates retrieval-heavy educational tasks from coordination-heavy ones. Second, it outlines a framework for comparing classical RAG, agentic RAG, and non-RAG multi-agent systems under shared budgets, source corpora, and evaluation protocols. Third, it identifies a benchmark and dataset strategy for evaluating architectural tradeoffs in educational AI without overstating downstream learner outcomes. The goal is not to argue that more agents are always better, but to clarify when orchestration complexity is justified and when classical retrieval-centric pipelines remain the right design choice.

1.1 Research Questions

The paper is organized around the following research questions.

RQ1: Under what educational task conditions is classical RAG sufficient as the primary orchestration pattern?

RQ2: When do agentic or multi-agent systems provide measurable advantages over classical RAG in pedagogical quality, adaptivity, or workflow completion?

RQ3: When retrieval is removed entirely, which classes of educational tasks degrade most severely, and which remain tractable through orchestration alone?

RQ4: What evaluation protocol most fairly separates gains from retrieval, gains from orchestration, and gains from additional budget?

1.2 Hypotheses

To keep the framing falsifiable, the paper adopts balanced hypotheses rather than a universal superiority claim.

H1: On low-coordination, knowledge-grounded educational tasks, classical RAG will perform comparably to or better than more complex multi-agent systems at lower cost.

H2: On coordination-heavy educational tasks such as adaptive tutoring, rubric-based critique, and multi-step study planning, agentic or multi-agent systems may outperform classical RAG on pedagogical quality and adaptivity.

H3: Non-RAG multi-agent systems will underperform retrieval-enabled systems on fact-intensive or citation-sensitive tasks, showing that orchestration alone does not remove the need for grounding.

H4: A plausible hypothesis for mixed-regime educational tasks is that hybrid systems will prove strongest: multi-agent orchestration with retrieval available as a conditional tool rather than as the sole pipeline pattern.

2 Background and Related Work

The literature relevant to this paper falls into four partially overlapping threads. The first is classical retrieval-augmented generation, especially educational assistants that rely on course materials, textbooks, lecture notes, or institutional knowledge bases. The second is agentic RAG, where retrieval is embedded inside iterative planning, tool use, critique, or workflow-based generation. The third is the broader literature on multi-agent large language model systems, including role-specialized architectures for decomposition, verification, and coordination. The fourth is the literature on intelligent tutoring, pedagogical dialogue, feedback generation, and adaptive educational assistants [3, 7]. Taken together, these threads motivate the paper's main distinction: the architectural question is not just whether retrieval is present, but whether retrieval is the dominant controller of behavior.

Classical RAG remains the default architecture for knowledge-grounded assistance because it is simple, interpretable, and often effective on factual question answering or source-grounded explanation. That makes it a strong baseline rather than a straw man. At the same time, recent agentic RAG systems argue that retrieval quality alone is not enough when tasks require decomposition, retrieval refinement, or iterative verification. The multi-agent RAG literature is especially relevant because it offers both positive and negative evidence: some systems improve hard multi-hop or retrieval-sensitive tasks through coordination, while other evaluations show that coordination overhead can underperform a strong classical RAG baseline in narrower factual settings [10, 12].

The key analytical distinction for this paper is therefore not simply whether a system uses retrieval, but whether retrieval is the dominant architectural pattern or one tool inside a broader orchestration strategy. Existing work often mixes these categories. Some systems are essentially classical RAG with stronger prompts or reranking. Others perform iterative retrieval and self-critique. Still others use multiple role-specialized agents that may or may not call retrieval depending on task demands. A careful taxonomy is needed so that comparisons do not collapse genuinely different design choices into a single "agentic" bucket.

The educational literature reinforces this need for separation. Education-facing benchmarks such as EduBench, TutorEval, ScienceQA, MathTutorBench, and TutorBench emphasize pedagogical quality, adaptivity, explanation, and instructional alignment rather than only factual correctness [18, 3, 8, 9, 15]. By contrast, transferable benchmarks such as HotpotQA, SCROLLS, LongBench v2, FEVER, Wizard of Wikipedia, BEIR, and AgentBench isolate mechanism-level abilities such as retrieval depth, evidence justification, long-context handling, grounded dialogue, retrieval robustness, fact verification, and workflow coordination [19, 14, 2, 17, 4, 16, 6]. The paper treats these two families differently: education-specific benchmarks support claims about educational usefulness, while transferable benchmarks support narrower claims about retrieval, reasoning, or orchestration mechanisms.

This section therefore serves two purposes. It situates the proposed taxonomy against existing architectures, and it clarifies what current benchmarks can and cannot justify. This matters because the paper's contribution is not to claim that multi-agent systems already dominate across education, but to provide a defensible framework for comparing classical RAG, agentic RAG, and non-RAG multi-agent systems under the right task and benchmark regimes.

Task regime	Grounding need	Coordination complexity	Hypothesized strong architecture
Evidence-grounded educational QA	High	Low	Classical RAG
Source-grounded explanation	High	Low to medium	Classical or agentic RAG
Rubric-based feedback	Medium	Medium to high	Agentic RAG or multi-agent orchestration
Adaptive tutoring dialogue	Low to medium	High	Multi-agent orchestration or agentic RAG
Lesson or study planning	Low to medium	High	Multi-agent orchestration

Table 1: Working task taxonomy for comparing retrieval-centric and coordination-centric educational AI architectures. The labels identify hypotheses to be tested against benchmark evidence rather than completed findings.

3 Framework

We define educational AI tasks along two axes. The first axis is *knowledge grounding need*: the extent to which success depends on retrieving accurate, task-relevant evidence from an external corpus or tool. The second axis is *pedagogical coordination complexity*: the extent to which success depends on decomposition, adaptive sequencing, feedback refinement, critique, learner modeling, or role-specialized reasoning. The value of the framework is not that it predicts winners in advance with certainty, but that it forces architectural comparisons to state which difficulty is actually being tested.

Under this framework, classical RAG is expected to remain strong in the low-coordination, high-grounding regime. Examples include factual question answering over course materials or evidence-grounded explanation tasks with limited personalization. By contrast, as pedagogical coordination complexity increases, systems may benefit from explicit orchestration. Agentic RAG represents one intermediate design in which retrieval is still available but is no longer the sole driver of system behavior. Non-RAG multi-agent systems test a different hypothesis: whether orchestration alone can solve some educational tasks without a dedicated retrieval pipeline. These are working hypotheses rather than settled conclusions, and the paper's evaluation protocol is designed specifically to expose where they fail.

The framework also encourages explicit role definition. Candidate roles include planner, retriever, tutor, critic, rubric-checker, misconception diagnostician, and safety or policy monitor. Not every task needs every role, and a key part of the contribution is to identify which roles matter under which task regimes. The final paper will specify how role specialization, communication, memory, and tool access interact, along with the design tradeoffs they introduce in latency, token cost, and failure surface.

Table 1 summarizes the paper's working taxonomy. The table is intended as an operational hypothesis that will later be stress-tested against benchmark results; it is not a declaration that any architectural family has already won a regime.

4 Evaluation Design

This section defines an evaluation protocol for a future comparative study; it does not report completed experimental results. The role of the protocol is to translate the taxonomy into a fair

Regime	Representative learning activity	Retrieval emphasis	Coordination emphasis
Evidence-grounded reasoning	Explain why the answer is correct with citations or reference material	High	Moderate (organizing retrieved context)
Rubric-based feedback	Provide diagnostic comments on student writing or code using a rubric	Moderate	High (pedagogical framing, tone, and scaffolding)
Adaptive tutoring	Diagnose misconceptions and offer next-step hints or mini-lessons	Low	High (memory of student state, strategy planning)
Lesson/exercise planning	Produce sequences of learning activities aligned to goals	Low	Very High (task decomposition, role assignment, pacing)

Table 2: Task regimes that divide retrieval-heavy experiments from coordination-heavy ones.

comparison among architectural families while keeping educational claims, mechanism claims, and cost claims distinct.

The architectural families are operationalized as follows. *Classical RAG* uses a single retrieve-then-generate pipeline with fixed retrieval access and no iterative planning loop. *Agentic RAG* preserves retrieval but allows iterative planning, critique, rerouting, or tool use around the same grounding layer. *Non-RAG multi-agent systems* rely on multiple specialized agents without a dedicated retrieval pipeline as the default controller, though lightweight lookups may still be possible depending on implementation. An optional *single-agent no-RAG* baseline helps isolate retrieval and coordination effects from general model quality.

The evaluation strategy is built around architectural fairness, boundary-condition testing, and a skeptical reading of output quality. The goal is to establish where retrieval-heavy conditioning is decisive, where coordination and pedagogical orchestration matter more, and where neither earns a clear superiority claim. Classical RAG, agentic RAG, and non-RAG multi-agent systems remain distinct families throughout the study; the optional single-model non-RAG baseline simply anchors the comparison so that retrieval and coordination gains can be disentangled. To avoid budget misattribution, every condition reports token usage, tool calls, and wall-clock latency, and differences in runtime cost are discussed alongside accuracy-like metrics.

Task regimes are separated explicitly: retrieval-heavy tasks (evidence-grounded question answering, factual explanation, citation-aware justification) are evaluated independently from coordination-heavy regimes (adaptive tutoring, misconception diagnosis, lesson sequencing, rubric-based feedback). This separation keeps the analysis from collapsing architectural differences into a single aggregate score. Table 2 summarizes the regimes, their dominant evaluation concerns, and the orchestration behaviors they stress.

The benchmark shortlist is produced by three filters. Each candidate must (1) map cleanly to one of the regimes in Table 2, (2) allow a contrast between at least two of the architectural families, and (3) surface metrics or human-evaluation protocols that clarify why an architecture performed better in that regime. Benchmarks that reward only short-answer factual correctness are retained only for the retrieval-heavy column; the coordination-heavy column relies on rubric-based feedback, tutoring dialogues, or planning tasks where pedagogy, not just accuracy, drives differentiation. The point is not to maximize benchmark count, but to make each benchmark legible as evidence for a specific kind of claim in the planned study.

The current shortlist is grounded in benchmark families that already expose the contrast

Benchmark	Category	Main task family	Main metrics	Comparison utility	Limitation for paper claims
EduBench	Education-specific	Multi-scenario educational QA, grading, and pedagogical support	12 dimensions covering adaptation, accuracy, and pedagogy	Compares factual grounding with personalization	Synthetic judges; best suited for educational claims rather than classroom-effect claims
TutorEval / LM-Science-Tutor	Education-specific	Open-book vs. closed-book science tutoring QA	LM-based key-point coverage with open- and closed-book settings	Contrasts retrieval-sensitive tutoring with agentic explanation strategies	Science-specific and evaluator-dependent
ScienceQA	Education-specific	Multimodal science QA with explanations and lecture context	Subject/context accuracy plus explanation scoring	Focuses on explanation quality and grounding within science domains	Not a full tutoring-dialog benchmark
MathTutorBench / TutorBench	Education-specific	Tutoring dialogue that includes scaffolding or rubric-based feedback	Accuracy plus pedagogy/rubric metrics	Highlights coordination-heavy pedagogy and instructional quality	Subject- or rubric-specific; generalization is limited
HotpotQA	Transferable	Multi-hop open-domain QA with supporting facts	Answer EM/F1, supporting fact EM/F1, joint EM/F1	Stresses retrieval depth and justification in classical vs. agentic retrieval	Supports mechanism claims, not pedagogical claims
AgentBench	Transferable	Tool-using workflow completion and coordination	Task success rates across environments	Emphasizes coordination and tool-use stress testing	Not education-specific; reserve for orchestration claims
SCROLLS / LongBench	Transferable	Long-context reasoning, QA, and summarization	Task-specific accuracy or ROUGE-style metrics	Stress test for long-context retrieval planning	Not pedagogy-specific
BEIR / FEVER / Wizard of Wikipedia	Transferable	Retrieval robustness, fact verification, and grounded dialogue	nDCG/MRR/Recall, FEVER score, grounded-dialog metrics	Isolates retrieval and grounding quality before pedagogical claims	Supports retrieval-focused claims only

Table 3: Current benchmark mapping for the proposed evaluation protocol. The table identifies which benchmark families would support which claims in a future study: education-specific benchmarks support educational-usefulness claims, while transferable benchmarks support narrower mechanism-level claims.

this paper cares about and therefore serve as inputs to the proposed protocol rather than as evidence of completed comparisons. EduBench, TutorEval, ScienceQA, MathTutorBench, and TutorBench give coverage over pedagogical quality, explanation, tutoring, and feedback regimes [18, 3, 8, 9, 15]. HotpotQA, SCROLLS, LongBench v2, FEVER, Wizard of Wikipedia, BEIR, and AgentBench provide complementary stress tests for retrieval depth, evidence justification, grounded dialogue, long-context reasoning, retrieval robustness, fact verification, and workflow coordination [19, 14, 2, 17, 4, 16, 6].

Table 3 gives the current benchmark mapping supported by the literature review. The mapping is preparatory: it specifies which benchmark families would justify which claims in a future comparative study. It intentionally separates *education-specific* benchmarks, which can support education-facing claims, from *transferable* benchmarks, which support narrower mechanism-level claims about retrieval, justification, long-context handling, or orchestration.

Evaluation combines automated metrics, human judgment, and cost reporting. Automated families cover groundedness, citation density, task completion, and robustness indicators such as failure-mode counts or hallucination rates. Human evaluation focuses squarely on pedagogical

Metric family	Reporting focus
Groundedness	Citation recall/precision, hallucination counts, retrieval coverage, and alignment between retrieved snippets and claims.
Pedagogical quality	Human ratings for specificity, scaffolding, level alignment, clarity of next steps, and sensitivity to learner state.
Adaptivity	Evidence of maintaining student context, handling follow-up prompts, and adjusting to rubrics or learner models mid-dialogue.
Cost and latency	Token usage, tool call counts, wall-clock latency per turn, and any orchestration overhead (planner cycles, critique rounds).
Robustness	Failure-mode counts, rerun stability, and robustness to input perturbations (e.g., misspelled prompts, conflicting goals).

Table 4: Metric families that keep groundedness, pedagogical quality, adaptivity, and cost separate.

Family	Retrieval	Orchestration style	Main purpose
Classical RAG	Enabled	Single retrieve-then-generate pipeline	Serves as a strong baseline for evidence-grounded tasks
Agentic RAG	Enabled	Iterative planning, critique, and tool-use control loops	Tests whether orchestration around retrieval improves grounding
Non-RAG multi-agent	Disabled or minimal	Multiple specialized agents coordinating their reasoning	Probes whether coordination alone can handle coordination-heavy educational tasks
Single-agent no-RAG	Disabled	Single model without retrieval orchestration	Provides a sanity anchor for interpreting retrieval and coordination gains

Table 5: Architectural comparison families for the proposed evaluation.

quality: specificity, scaffolding, actionable steps, level appropriateness, adaptivity to the described learner, and rubric-aligned critique when applicable. Human raters also flag whether they believe the system output would plausibly influence the next instructional move; this human-centered lens helps remind readers that the paper evaluates system outputs, not real learner outcomes. Any claim about learner progress is explicitly framed as a hypothesis that would require a longitudinal study, so the paper remains rigorous about the gap between output quality and actual classroom learning.

Metric families are summarized in Table 4, showing that cost, adaptivity, groundedness, and pedagogical quality are tracked separately so no single blended score hides boundary conditions.

Table 5 revisits the baseline families, clarifying that classical RAG remains the default retrieval-heavy pipeline, agentic RAG adds orchestration around the same retrieval layer, and non-RAG multi-agent systems rely on coordination rather than retrieval. The optional single-agent no-RAG baseline provides an additional control point for understanding whether retrieval or coordination explains gains.

4.1 Benchmark Selection Rationale

We document the benchmark filters so reviewers can trace which regimes are represented. The shortlist also notes dataset limitations, such as narrow domain coverage or question templates, and the exact prompts used to guarantee task parity. If certain regimes lack robust public benchmarks,

we describe the synthetic or controlled setups that approximate the regime, such as pairing retrieval prompts with tutoring-style instructions, and highlight the resulting boundary conditions.

One important rule governs interpretation: education-specific benchmarks support claims about pedagogical usefulness or instructional quality, while transferable benchmarks support only narrower claims about retrieval, evidence justification, long-context handling, or orchestration. This prevents the paper from overextending general agent benchmarks into classroom claims they cannot support, even when those general benchmarks are informative about the mechanisms behind agentic retrieval or multi-agent coordination [16, 6].

4.2 Fairness Controls

Reviewers will demand proof that comparisons are fair. The following controls are in place:

- **Baseline strength:** the classical RAG condition uses well-tuned retrieval depths, reranking, and answer fusion so it cannot be dismissed as a straw man.
- **Budget transparency:** token usage, tool calls, planner or critic cycles, and per-turn latency are logged for every architectural family, making it possible to ask whether a win is due to more compute or fundamentally better orchestration.
- **Corpus parity:** whenever retrieval is available, every system sees the same document set and obeys identical indexing, chunking, and retrieval-budget policies.
- **Task parity:** each system receives the same prompt wording, student profile metadata, and rubric anchors when applicable, so differences reflect architecture rather than prompt engineering.
- **Human evaluation rigor:** annotator instructions differentiate between output quality and learner-outcome claims, and inter-rater calibration focuses on pedagogical judgments rather than superficial fluency.
- **Metric separation:** groundedness, pedagogical quality, adaptivity, cost, and robustness are reported independently, preventing a "one number to rule them all" interpretation.

4.3 Reviewer Objections and Ablations

We anticipate several predictable concerns and plan ablations to address them. One ablation disables coordination layers in agentic RAG, testing whether retrieval plus orchestration yields the same gains as orchestration plus retrieval. Another adds retrieval to the strongest non-RAG multi-agent baseline to show whether grounding alone could rescue coordination failures. A third reduces tool-call budgets to confirm that reported wins are not simply artifacts of unlimited compute. These ablations double as boundary-condition tests: if an orchestrated agentic RAG system performs at parity with classical RAG on evidence-heavy tasks, the protocol is designed to document the cost delta; if non-RAG multi-agent systems fail on citation-sensitive tasks, the protocol is designed to record those failure modes. Mixed or negative outcomes are therefore informative because they may expand the domain where classical RAG remains the preferred default.

5 Discussion

This paper is not yet an empirical verdict, and it should not be read as a universal replacement story. The more defensible contribution is a decision framework for educational AI architecture

selection: classical RAG may remain the right top-level abstraction for straightforward evidence-grounded tasks, while coordination-heavy educational workflows may justify agentic or multi-agent orchestration.

This perspective has practical implications. It suggests that educational systems should be designed with explicit awareness of whether their failures are caused by missing evidence, weak orchestration, poor pedagogical adaptation, or some combination of all three. It also suggests that the most capable systems may be hybrid ones in which retrieval is preserved as an internal tool while orchestration logic is handled by planners, critics, or specialized pedagogical roles.

The discussion must also foreground limitations. Current educational benchmarks may not fully capture authentic tutoring quality, learner modeling, or long-term instructional effectiveness. Multi-agent systems also introduce higher cost, more complicated debugging, and new coordination failure modes. These limitations narrow the scope of the claims and reinforce why the paper is framed around architecture selection and evaluation design rather than broad classroom-impact claims.

An important meta-point is that negative or mixed results are still informative. If classical RAG remains best on simple educational tasks, that strengthens the paper's taxonomy rather than undermining it. Likewise, if future evaluations show that retrieval-enabled multi-agent systems outperform non-RAG systems on fact-heavy tasks, that would support a hybrid conclusion: the main architectural shift is not from retrieval to no retrieval, but from a rigid retrieve-then-generate pipeline to conditional orchestration with retrieval as one available capability.

Fairness controls and cost transparency are central to these claims. The evaluation design reports token usage, tool calls, planner rounds, and latency so that readers can judge whether an architectural family earns wins through better pedagogy or simply through greater compute budgets. Similarly, human evaluation protocols are structured to distinguish output quality, such as fluency, citation density, and scaffolded hints, from speculative claims about learner gains. Even a hybrid conclusion should therefore be read as contingent on future human evaluation under the proposed protocol, not as something already established by the current draft. When reviewers demand evidence of pedagogical quality, the paper can point to these controls and to the adaptivity and cost metrics that keep the conversation grounded in observable behaviors rather than imagined classroom effects.

Finally, boundary conditions are critical to the decision framework. The proposed evaluation is designed to document where classical RAG still dominates, such as evidence-heavy prompts with tight citation requirements, where agentic RAG adds a useful coordination layer, and where non-RAG multi-agent systems break down because they lack grounding. By treating orchestration-heavy regimes separately from retrieval-heavy ones and by reporting the associated failure modes, the discussion makes it clear that no family should be assumed to enjoy universal superiority, only context-dependent relevance.

6 Conclusion

Educational AI systems are often evaluated as if retrieval quality alone determines usefulness. This draft argues for a broader view. The relevant design question is not simply whether to retrieve documents, but whether classical retrieve-then-generate pipelines are sufficient for tasks that demand pedagogical sequencing, critique, adaptation, and role specialization. A rigorous comparison among classical RAG, agentic RAG, and non-RAG multi-agent systems can clarify when retrieval should remain central, when orchestration should dominate, and when hybrid designs are most justified.

The final paper aims to contribute both a conceptual taxonomy and a practical evaluation

protocol, along with a benchmark mapping that distinguishes education-facing evidence from transferable mechanism-level evidence. Its strongest version will not claim that multi-agent systems universally outperform retrieval-based approaches. Instead, it will identify the conditions under which educational assistance may require more than classical RAG and the conditions under which classical RAG should remain the preferred default. In that sense, the paper's contribution is less a verdict than a disciplined way to ask the right comparative question.

A Benchmark and Dataset Mapping

This appendix summarizes the benchmark families currently identified for the paper's evaluation blueprint. The list is intentionally split into *education-specific* benchmarks and *transferable* benchmarks so that claim scope remains disciplined.

A.1 Education-specific benchmarks

- **EduBench** [18]: a broad educational benchmark spanning student- and teacher-facing scenarios with twelve pedagogically oriented dimensions. It is useful for testing whether an architecture improves pedagogical quality and personalization rather than only factual correctness.
- **TutorEval / LM-Science-Tutor** [3]: a science tutoring benchmark with open-book and closed-book conditions. It is especially useful for comparing retrieval-enabled tutoring against more agentic explanation strategies.
- **ScienceQA** [8]: a multimodal science QA benchmark with explanations and lectures. It is useful for explanation quality and grounding, but does not fully capture tutoring dialogue or adaptive pedagogy.
- **MathTutorBench / TutorBench** [9, 15]: tutoring and feedback-oriented benchmarks that place more weight on scaffolding, rubric adherence, or pedagogical quality than on factual QA alone.

A.2 Transferable benchmarks

- **HotpotQA** [19]: a strong stress test for multi-hop retrieval and evidence justification.
- **AgentBench** [6]: a coordination and tool-use stress test for workflow-oriented agents.
- **SCROLLS / LongBench** [14, 2]: long-context stress tests relevant to lecture notes, textbooks, and curriculum-scale materials.
- **BEIR / FEVER / Wizard of Wikipedia** [16, 17, 4]: retrieval, verification, and groundedness benchmarks useful for isolating retrieval quality from pedagogy.

A.3 Interpretation rule

Education-specific benchmarks can support claims about pedagogical quality, instructional usefulness, or adaptive support. Transferable benchmarks can support only narrower claims about retrieval depth, evidence justification, long-context reasoning, or coordination. This distinction is essential to prevent the paper from overclaiming educational significance from general-purpose agent benchmarks.

A.4 Evaluation harnesses and reusable infrastructure

The current repo search identified reusable infrastructure that could support later empirical work, including MASEval for adapter-based agent evaluation, HAL Harness for benchmark orchestration across agentic tasks, a lightweight RAG evaluation harness for retrieval metrics, and educational grading infrastructure such as Autolab [5, 11, 13, 1]. These artifacts strengthen the practical feasibility of the proposed evaluation protocol, but they do not by themselves justify educational claims.

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